



CHAPTER

Telemetry and Remote Connectivity

Router, remote-access gateway and cloud monitoring

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16.1 Overview and function

A cellular router and cloud dashboard give operators and HydroVolta service remote visibility into the unit's process values and alarms, independent of the local HMI (chapter 14).



Figure 16.1. Suggested figure: photograph of the router/gateway mounted in the cabinet, with antenna and SIM slot visible.

16.2 Key specifications

- **Confirmed deployed remote access:** real OpenVPN client configuration files exist in HydroVolta's engineering library for at least two pilot containers (Farys, BASF/CAGECE demo), following a router-based “meeting point” VPN architecture (Weidmüller IE-SR-2GT-LAN / IE-SR-2GT-UMTS routers, per the vendor tech note on file) — this is the one telemetry element confirmed as *actually used*, not just evaluated.
- **Confirmed deployed dashboard: ThingsBoard**, with a real MQTT parameter list and full tag list captured from the BASF pilot deployment.
- A real 2023 requirements-workshop record (source: 08_IT/00_OQuila progress) describes an **actually running 4-container pilot fleet** at that time: **Advantech ADAM-3600** connecting directly to **Azure IoT Hub**, piped through **Azure Stream Analytics** into a SQL database and visualized in **Power BI**; connectivity via a **4G Teltonika router**, with HydroVolta noting a possible future switch to Ewon “for a better SLA” due to past Teltonika stability issues. Each container sent a telemetry message roughly every 10 seconds ($\approx 8,000$ messages/container/day: connectivity, pressure, flow, temperature, energy). Parameter changes at that time were only possible via VPN, not from the cloud dashboard.
- **OQuila** was a third-party IoT consultancy HydroVolta held a 2023 requirements workshop with, to evaluate a longer-term telemetry platform (long-term storage, a graphical P&ID front-end, configurable alarm thresholds, role-based access). HydroVolta's own workshop notes state ThingsBoard was assessed as “ok” for the dashboard need on its own. **No document confirms OQuila's proposal was ultimately adopted or built** — treat the Azure IoT Hub/Power BI pipeline above and the ThingsBoard/ Weidmüller-OpenVPN pipeline elsewhere in this manual as two telemetry architectures documented at different points in HydroVolta's history, not necessarily the same one. TODO: confirm which architecture (if either, unchanged) is used on the current standard SalinBloc™ product.
- Controller reached over the network: Weidmüller UC20-WL2000-AC (chapter 14).

16.3 Operation

Process values available on the cloud dashboard mirror those on the local HMI (chapter 6). TODO: no document describes ThingsBoard dashboard login/navigation for an end customer.

16.4 Maintenance

TODO: no router/SIM maintenance procedure (data plan renewal, firmware update schedule) was found.

16.5 Troubleshooting

- **Router unreachable:** confirm the device is powered and check its cellular signal/SIM status before assuming a software fault.
- **Cloud dashboard shows stale data:** check the router's connection to the controller first; if the controller itself is reachable locally but not remotely, the fault is likely in the router/network path rather than the controller.

See also the ADAM-3600 and Modbus-RTU troubleshooting steps in chapter 14, which apply to field-device connectivity issues more broadly.